

Jingwen (Lisa) Zhong

lisazjw.zhong@mail.utoronto.ca | 647-269-3189 | <https://lisazhong13.github.io/>

EDUCATION

MSc in Applied Computing, Data Science Concentration

Toronto, ON

University of Toronto | **Vector Scholarship in Artificial Intelligence (\$17,500)**

Sep 2026 – Jan 2028 (Expected)

- **Relevant Courses:** Neural Networks and Deep Learning, Computer Imaging, Data Science, Natural Language Computing, Statistical Methods for Machine Learning and Data Mining

Honours Bachelor of Science in Data Science Specialist & Actuarial Science Major | GPA 3.92/4.0

Toronto, ON

University of Toronto | **Dean's List; UofT Excellence Award 2024 & 2025 (\$15,000)**

Sep 2022 – Apr 2026

- **Relevant Courses:** Machine Learning, Data Structure and Algorithm Design, Time Series, Probability & Statistical Inference

WORK EXPERIENCES

Research Assistant – Spatiotemporal Machine Learning and Geospatial Downscaling

May 2025 – Present

Department of Statistical Sciences, University of Toronto (Preprint on arXiv: <https://doi.org/10.48550/arXiv.2512.05139>)

- Developed an end-to-end **spatiotemporal deep learning** framework to downscale 25 years of MERRA-2 atmospheric data from ~50 km to ~7 km resolution, achieving **R² up to 0.94** across regional evaluations.
- Built large-scale **geospatial data and degradation pipelines** for NetCDF climate data, implementing regridding, resolution degradation, autocorrelation analysis, and distribution/outlier diagnostics to construct and validate realistic training inputs.
- Designed and evaluated **U-Net, transfer-learning, and diffusion-based (DDPM)** models in PyTorch, incorporating multi-step temporal context, topographic information, and auxiliary spatial features.
- Developed rigorous **temporal out-of-distribution evaluation** across regions and seasons, diagnosing generalization failures through spatial maps, distributional analysis, R²/RMSE/MAE, and conservation tests rather than relying solely on random train-test splits.
- Currently developing a next-generation **encoder–temporal processor–decoder** framework with multi-scale objectives and Transformer-based temporal modeling to improve long-horizon generalization and uncertainty-aware high-resolution forecasting.

Data Science Intern

May 2026 – Aug 2026

Intact Financial Corporation

- Implemented a **reinforcement learning-based optimization framework** combining predictive modeling and decision optimization for large-scale pricing strategies.
- Developed and maintained model review and data collection workflows in **AWS**, using **GitHub** for version control and **Airflow** to automate multi-script execution and generate model diagnostic visualizations for anomaly detection.
- Built **PySpark** testing and analysis pipelines in **Databricks** to validate model outputs, business logic, and edge cases on large-scale datasets, improving model reliability and reproducibility.
- Implemented portfolio off-balancing algorithms in **Databricks**, querying production tables from **Snowflake** and applying Python-based optimization logic to satisfy global constraints while maintaining stable pricing behavior.

Actuarial and Data Science Intern

Jan 2026 – Apr 2026

Aviva

- Supported a major actuarial filing by preparing, validating, and reconciling policy, premium, and loss data across multiple source systems.
- Automated loss-data ingestion, transformation, and reconciliation workflows using **SAS, SQL, and Python**, integrating policy- and claim-level datasets across heterogeneous granularities.
- Built the model structure and reconciled premiums for the system migration of personal private trailer business, validating data consistency and investigating discrepancies between legacy and new systems.

Actuarial Analyst Intern

May 2025 – Aug 2025

Aviva

- Supported actuarial pricing by leveraging **SAS, SQL, and Earnix** to build analytical tools and validate models.
- Pre-processed and analyzed loss and exposure data through automated scripts in **Excel VBA** and **Python**, reducing data preparation time by 30% and streamlining pricing reviews.
- Improved model accuracy and consistency by reconciling discrepancies across data sources, eliminating 98% of data mismatches between variables and premium values.
- Designed and delivered broker-facing presentations on pricing changes, dislocation, and customer impact, effectively clarifying the rationale behind segmentation updates.

PROJECTS

CAS Case Competition – Hurricane Insurance Visualization (First Place).

Feb 2025

- Built full-stack web app (D3.js, Leaflet, PostgreSQL) for hurricane data exploration.
- Designed backend data pipeline for querying and visualization.

Kaggle PII Detection (Bronze Medal, Top 9%)

Feb 2024 – Apr 2024

- Fine-tuned DeBERTa-v3 model for named entity recognition on 7,000+ essays.
- Augmented dataset with 11,000+ synthetic samples using LLM APIs.
- Achieved 95.74% F1 score in large-scale competition.

SKILLS

- **Languages:** Python, SQL, Java, R, JavaScript, HTML/CSS, C
- **ML / Data:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
- **Data Engineering:** Spark, Databricks, Snowflake, Airflow, ETL
- **Tools:** AWS, Linux, Git, Power BI, Tableau